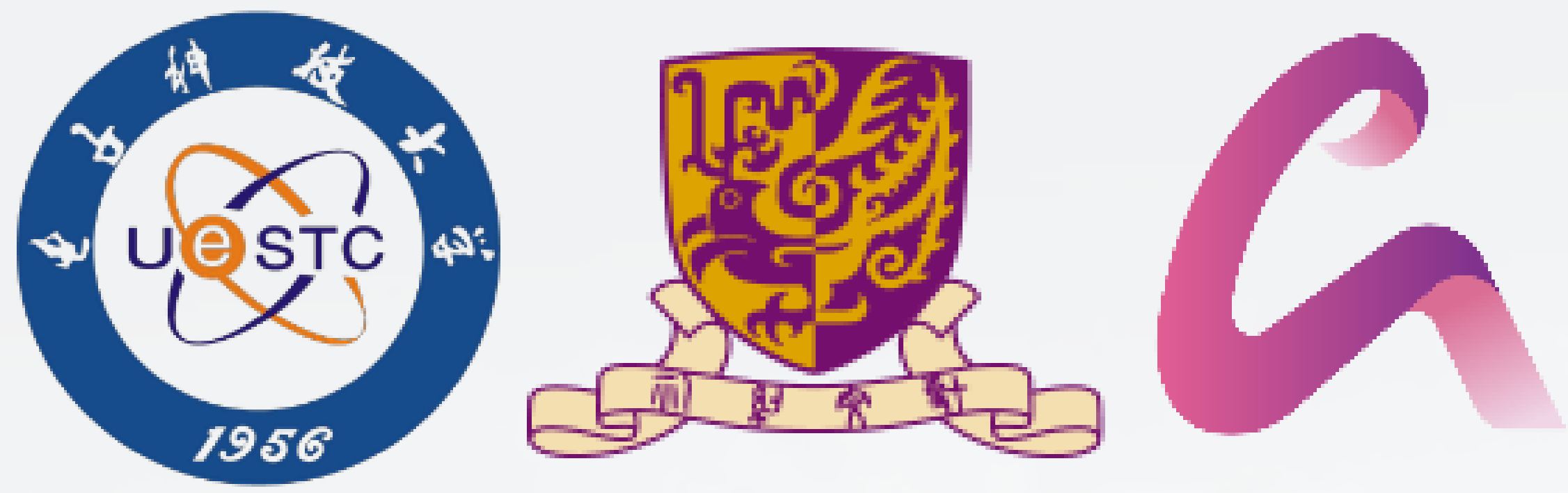


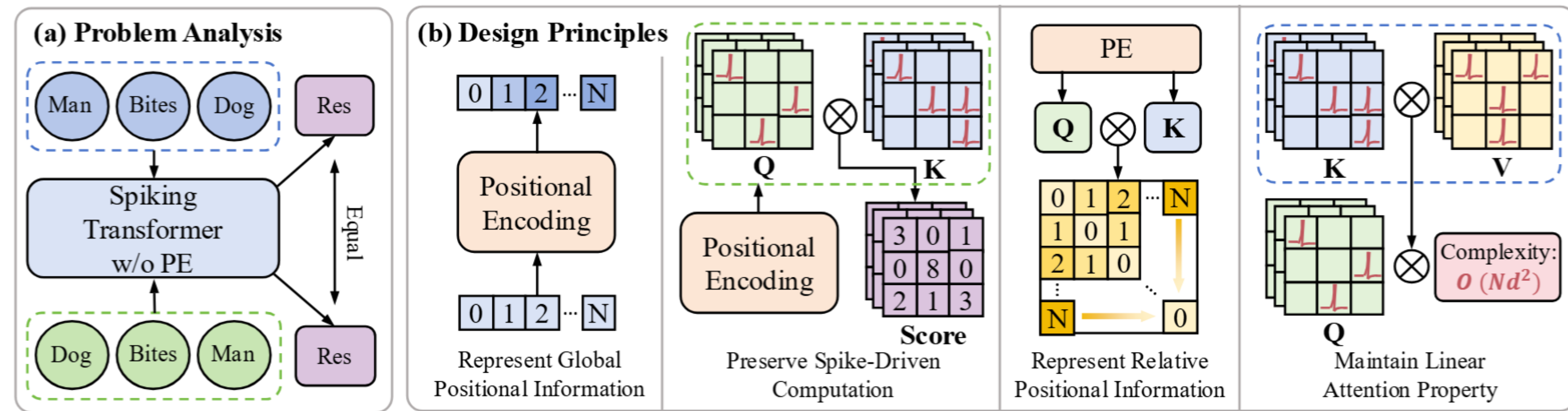


Positional Encoding for Spiking Transformers

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Motivation & Design Principles



- Positional information is essential for sequence modeling. However, in SSA-based Spiking Transformers, token reordering changes only the output order while leaving token representations unchanged, leading to permutation-invariant outputs.
- Existing ANN methods are not directly suitable for SNNs, as they may compromise spike-driven computation; moreover, absolute encoding lacks relative position modeling, while some relative encoding requires attention-map computation and breaks SSA's linear attention.
- An effective positional encoding for Spiking Transformers should **encode global positions**, **preserve spike-driven computation**, **capture relative positional relationships**, and **maintain SSA's linear complexity**.

Spiking Positional Encoding

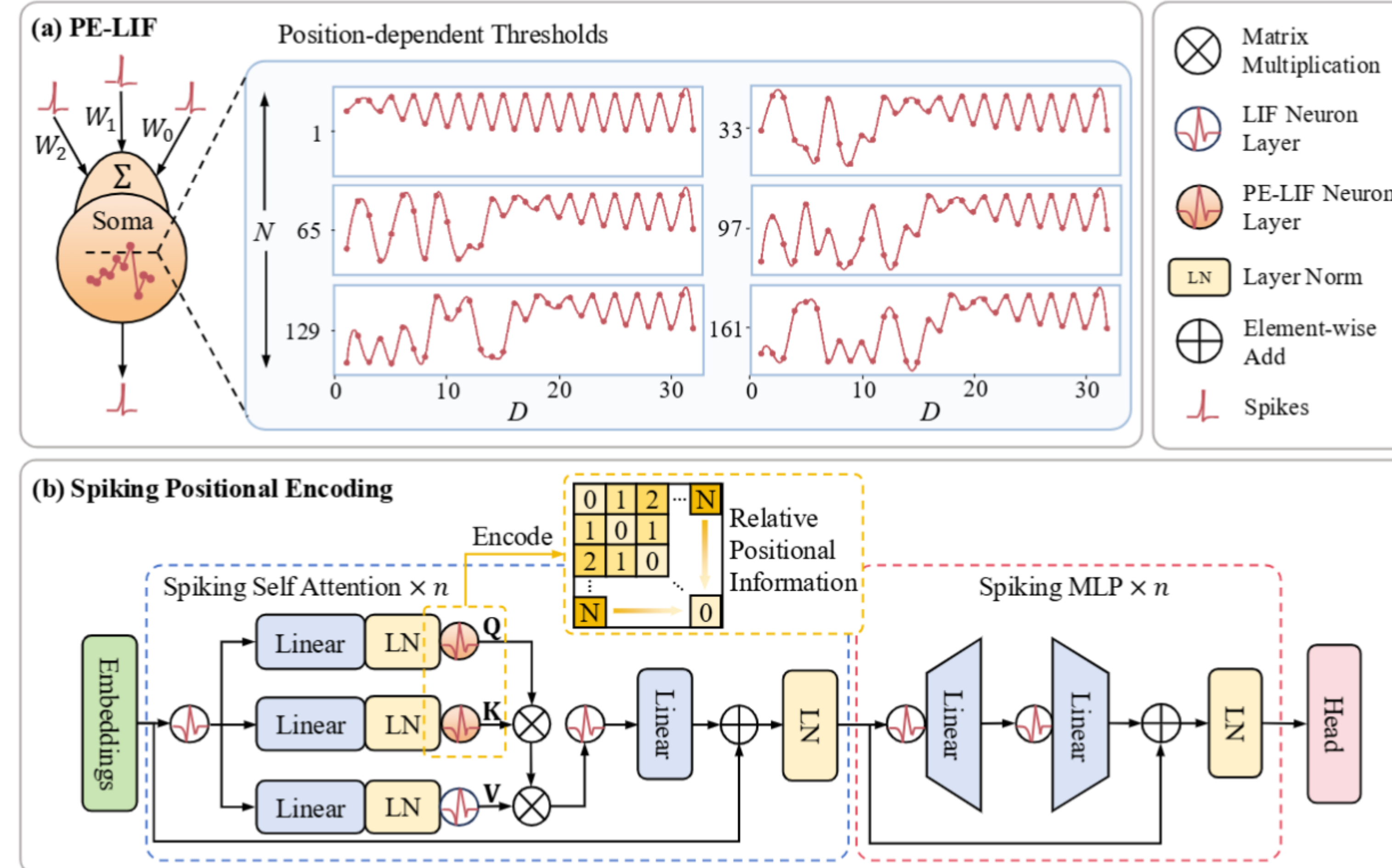
- We propose **Spiking Positional Encoding (SPE)**, which replaces the conventional LIF layers in the query and key branches of SSA with **Positional Encoding LIF (PE-LIF)** neurons.
- For a neuron at position i and dimension j , the threshold is defined as

$$\theta_{i,j} = \begin{cases} \theta + \lambda \cos\left(\frac{i}{10000 \frac{(j-1)}{D}}\right), & j \text{ is odd,} \\ \theta + \lambda \sin\left(\frac{i}{10000 \frac{(j-2)}{D}}\right), & j \text{ is even.} \end{cases}$$

- Thereby injecting global positional information directly into the spike-generation process.
- Since positional information is encoded through thresholds, PE-LIF still emits binary spikes $s_{i,j}(t) \in \{0,1\}$, preserving the spike-driven computation paradigm.
- By applying PE-LIF to $\mathbf{Q}$ and $\mathbf{K}$, SPE further encodes relative positional information into query-key interactions, as $\mathbb{E}[\mathbf{q}_m^T(t) \mathbf{k}_n(t)]$ contains a distance-dependent term $g(m-n)$.
- Because SPE modifies only the spike activation of $\mathbf{Q}$ and $\mathbf{K}$ without attention-map computation, it preserves the associative form $\mathbf{Q}(\mathbf{K}^T \mathbf{V})$ and maintains the linear attention property of SSA.

Spiking Positional Encoding (Continued)

- Moreover, $g(m-n)$ satisfies long-term decay, $\lim_{|m-n| \rightarrow \infty} g(m-n) = 0$, which suppresses spurious interactions between distant tokens and encourages the model to focus on positionally relevant dependencies.



- To enable SPE to encode relative positional information, the prerequisite $\mathbb{E}[\hat{u}_{i,j}(t)] = \mathbb{E}[s_{i,j}(t)]$ should be approximately satisfied.
- For this purpose, we introduce **Membrane Potential Regularization Surrogate Gradient (MPR-SG)** for the PE-LIF layers. MPR-SG keeps the forward output identical to the original binary spike, while implicitly induces an auxiliary regularization term, $\mathcal{L}_{Aux} \propto \ln \|\mathbb{E}_B[\mathbf{s}(t)] - \mathbb{E}_B[\hat{\mathbf{u}}(t)]\|_2$, thereby aligning membrane-potential statistics with spike statistics without explicit auxiliary losses or additional hyperparameters.

Experimental Results

GLUE benchmark

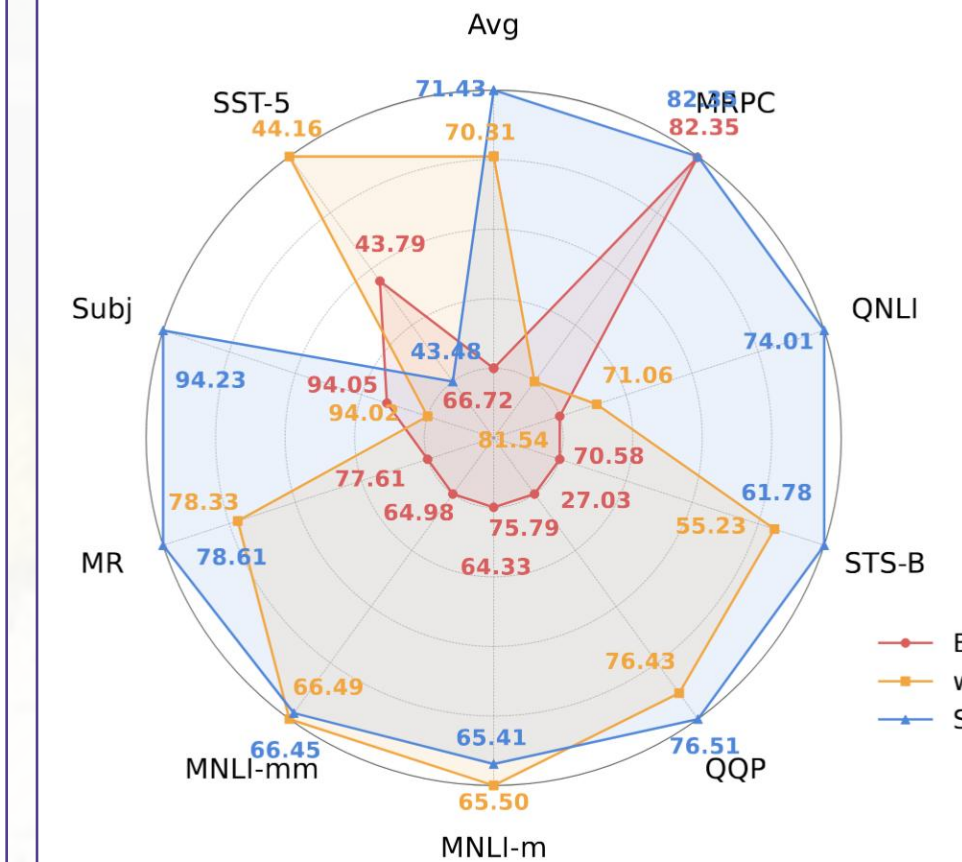
Model	Energy (mJ)	MNLI-m/mm	QQP	QNLI	SST-2	CoLA	STS-B	MRPC	RTE	Avg.
BERT _{base}	51.41	83.8/83.4	90.5	90.7	92.3	60.0	89.4	89.8	69.3	83.2
BERT _{3L}	12.9	77.1/77.1	85.2	85.8	88.1	31.7	85.7	86.4	66.4	75.9
ELMo	—	68.6/—	86.2	71.1	91.5	44.1	70.4	76.6	53.4	70.2
BiBERT	—	66.1/67.5	84.8	72.6	88.7	25.4	33.6	72.5	57.4	63.2
BiT	—	77.1/77.5	82.9	85.7	87.7	25.1	71.1	79.7	58.8	71.0
BiPFT	—	69.5/70.6	83.7	81.7	86.2	22.9	80.2	76.2	66.1	70.8
SpikeBERT	14.30	71.4/71.0	68.2	66.4	85.4	16.9	18.7	82.0	57.5	59.7
PSN-BERT	—	35.4/35.2	0.0	50.5	50.9	0.0	6.8	81.2	52.7	34.7
LIF-BERT	7.98	56.8/55.2	70.0	60.6	80.6	14.6	20.0	82.3	53.8	54.9
Spikingformer	6.76	71.9/72.5	84.7	76.0	87.2	24.4	54.5	79.7	55.6	66.8
Spikingformer w/ SPE	4.92	70.7/72.0	85.8	82.9	87.4	33.3	79.3	83.5	58.1	72.6

Experimental Results (Continued)

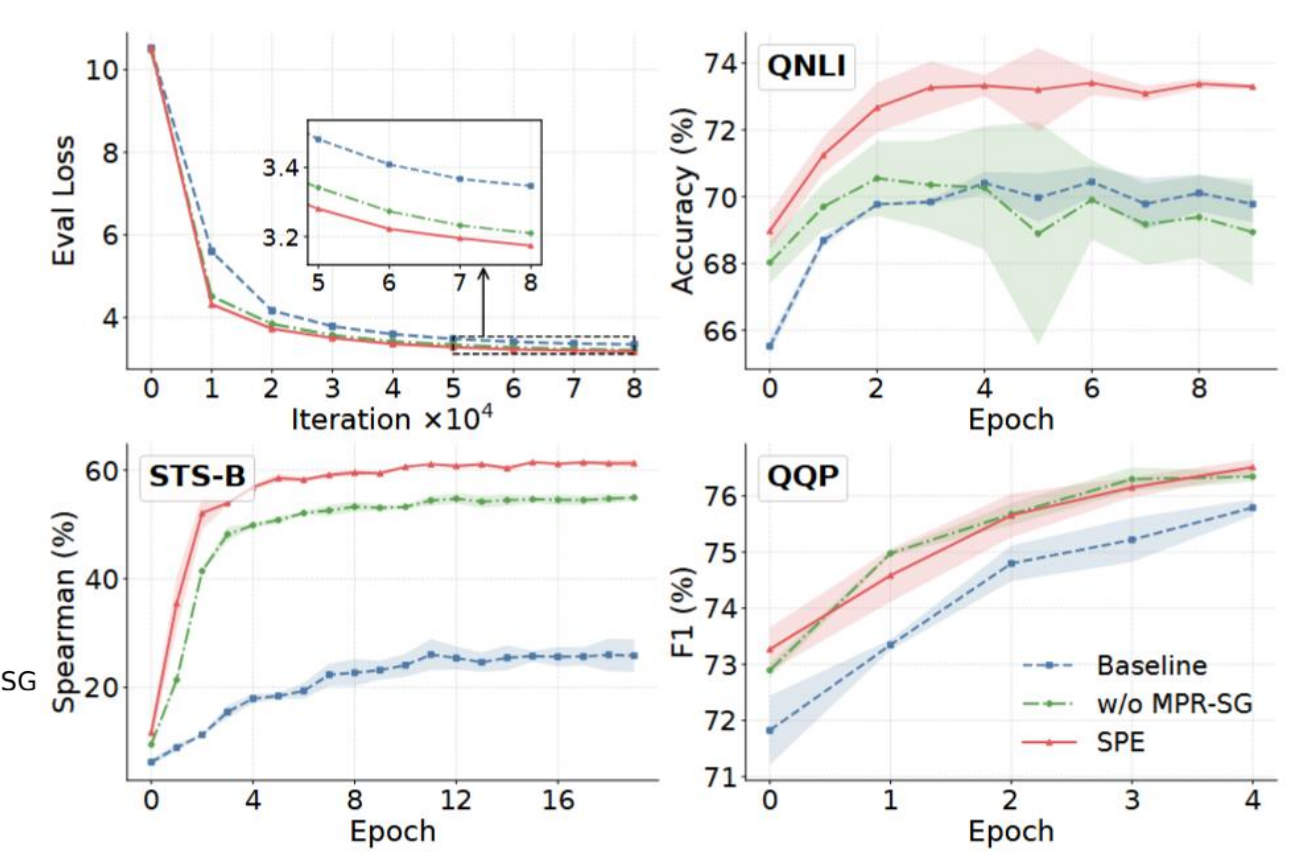
Comparison with other positional encoding methods

Model	Param(M)	Max Length = 128					Avg.	Max Length = 1024			Avg.
		MR	SST-2	Subj	SST-5			AGNEWS	IMDB		
BERT	109.8	87.63	92.31	95.90	50.41	81.56	94.50	92.10	93.30		
Spikformer w/o PE	109.8	75.87	81.71	91.60	41.84	72.76	—	—	—	—	—
Spikformer w/ CPG-PE	110.4	82.42	82.90	92.50	43.62	75.36	84.70	79.47	82.09		
Spikformer-XNOR w/o PE	109.8	75.80	81.74	91.50	41.88	72.73	—	—	—	—	—
Spikformer-XNOR w/ Gray-PE	109.8	83.73	84.52	92.50	44.06	76.20	84.92	79.79	82.36		
Spikformer w/ Log-PE	109.8	83.88	84.64	92.80	44.52	76.46	86.77	80.46	83.62		
Spikingformer w/ SPE	109.8	82.55	87.39	95.45	49.32	78.68	93.93	88.17	91.05		

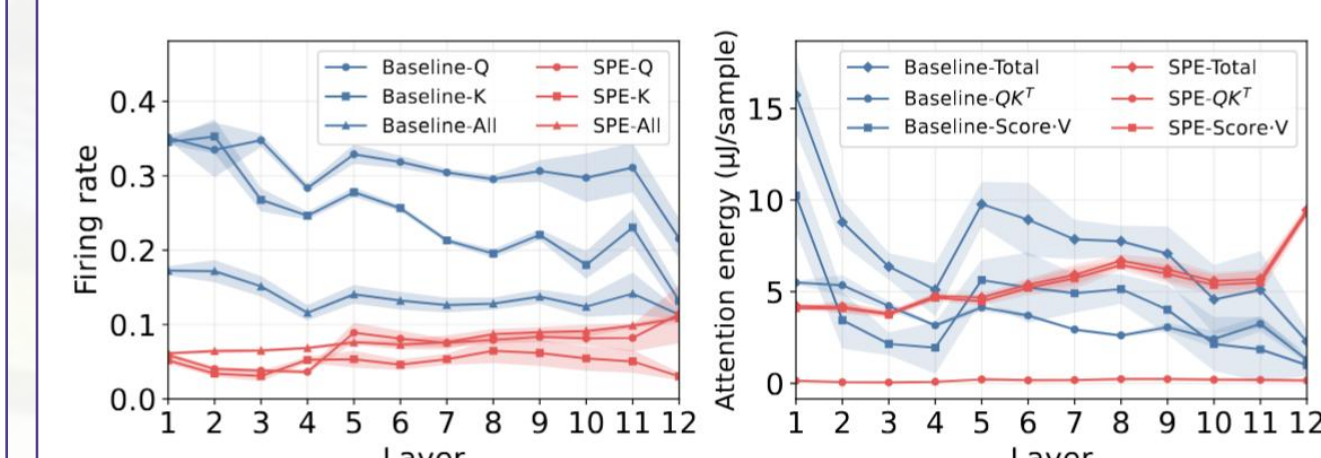
Ablation study



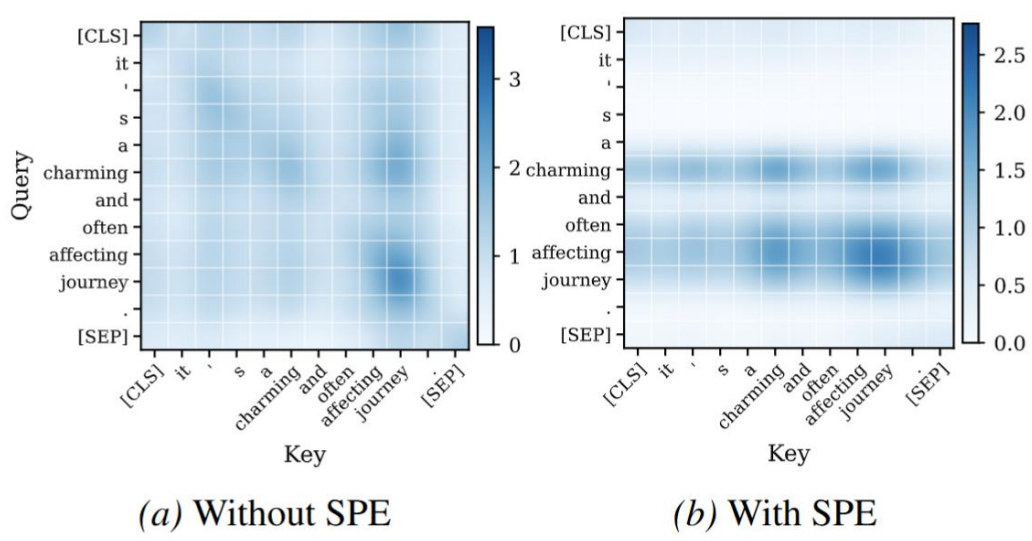
Rapid convergence



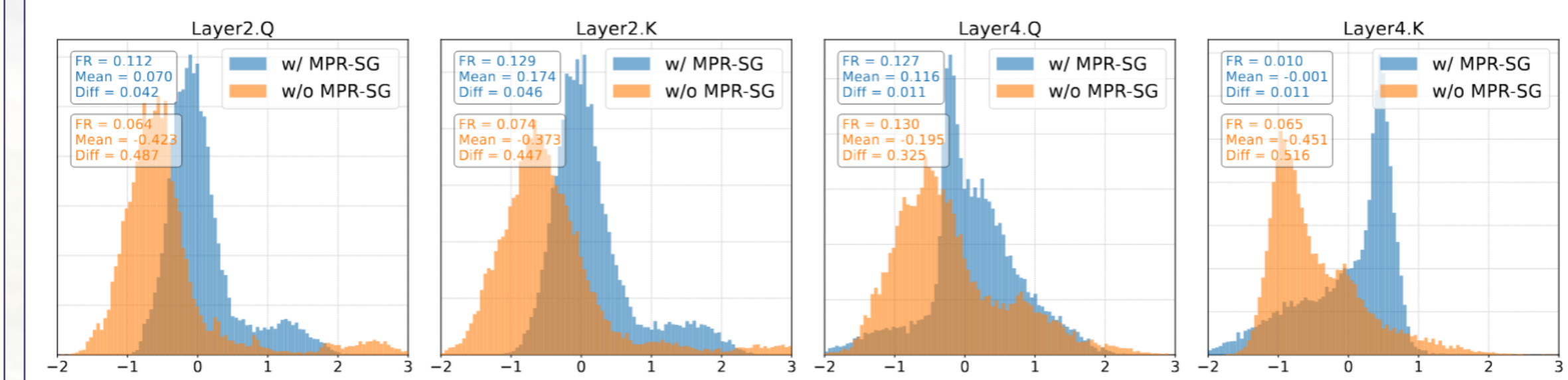
Energy reduction analysis



Token selectivity



Impact of MPR-SG



Conclusion

- We identify the necessity of positional encoding in Spiking Transformers and formulate four design principles for spike-driven sequence modeling.
- Guided by these principles, we propose SPE, which uses PE-LIF neurons to encode positional information through threshold modulation while preserving spike-driven computation.
- Theoretical analysis shows that SPE captures relative positional information and exhibits desirable long-term decay.
- Experiments on thirteen NLP datasets demonstrate that SPE consistently improves performance over existing positional encoding methods while maintaining the energy efficiency of SNNs.